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DAT321 DIT843
Basics in Software Quality

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(he/him)

Margaret Hamilton

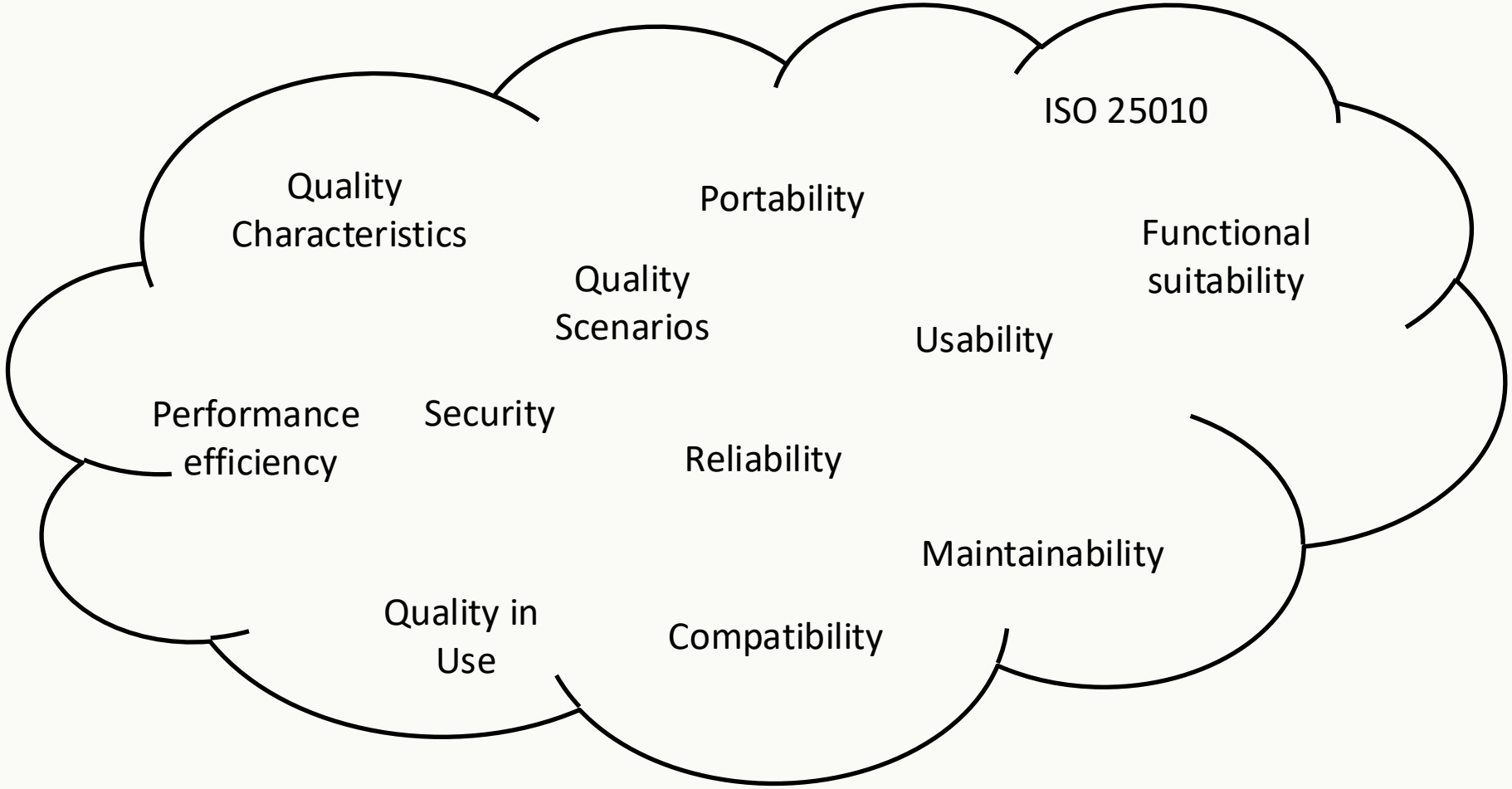
Coined the term Software Engineering

Leader of the development of on-board flight software for Apollo's missions

Presidential Medal of Freedom (2016), the highest civilian honor in the United States

Watch: https://www.youtube.com/watch?v=4sKY6_nBLG0

Today on QA and Testing



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What will we learn today?

- How to express **software qualities**? How to achieve those qualities?
- What is the **ISO 25010:2011**?
- What are the differences between **Product Quality** and **Quality in Use**?
- What are the **different quality** attributes in the ISO 25010?

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Warning! Abstract concepts ahead!

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Part 1:
Defining Quality

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Why Software Quality is Important?

- Software systems are pervasives and bad quality can lead to **hazardous situations**.
- Three examples of **tragic cases**: ☹
  - Therac-25: (80s)
  - Toyota unintended acceleration(1990s-2000s)
  - Boeing 737 Max (2018, 2019)

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Software quality: Why, What and How

- What is a **high-quality** software?
  - We all want to develop and use **high-quality software**.

- The problem in defining quality is that..
  - Definitions are **not congruent** and can be confusing.
  - Encompasses **many properties and behaviour** of the system.

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Quality Standards – ISO 25010:2011

- **ISO**: International Organization for Standardization
  - Develops and publishes worldwide technical, industrial and commercial **standards**.

- Those standards establish a **baseline of rules and artefacts** to foster congruent methods for measuring and assessing quality.
- We use the **ISO 25010:2011** for Software Quality

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From ISO 25010:2011:

"The quality of a system is **the degree** to which the system satisfies **stated and implied needs** of its various stakeholders, and thus provides **value**."

In other words: Make quality attributes **quantifiable and measurable**.

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What are important quality attributes?

You work for the Chat-Co, a company that offers a software product for chatting or messaging friends (think of WhatsApp, Messenger). Below we list important features about the software product and its development process:

F1: Each user account is connected to a single phone number. This number registered when creating the account.

F2: Besides text messages, users can send audio or video clips (limit of 3 minutes per message), or image files.

F3: Users can make video or audio calls with another user or a group chat that they are a part of. The user(s) on the other side of the call must accept the call before initiating the audio / video stream.

F4: The ChatApp can be used in mobile phones, tablets, or personal computers (PCs).

F5: The user interface must adapt, i.e., be responsive to the different layouts of screens from multiple types of devices.

F6: The development team does not have a fixed release calendar. New features and bug fixes are only added based on requests by the community of users.

F7: Users can customize their user interface by choosing different color themes, backgrounds, fonts, and text color in their chats.

F8: The ChatApp can import a list of contacts from the user's personal accounts (Google, Apple, Facebook, etc.).

Expressing Software Qualities

Distinguish between quality attributes and quality scenarios

- **Quality attributes** are the **characteristics** used to describe qualities.
  - Descriptive, high-level, abstract.
- **Quality scenarios** are the **application** of a quality attribute on a software system.
  - Concrete, testable, actionable.

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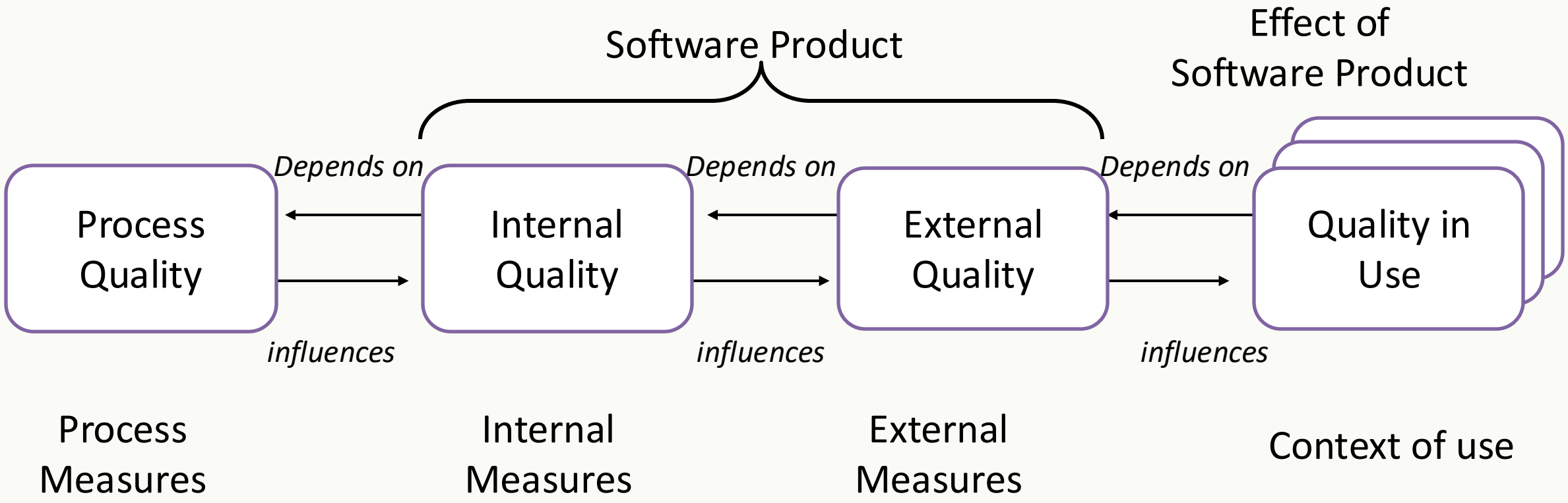
Different Types of Software Quality



Source: International Standard Organization, ISO/IEC 25010 standard

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Summary: Quality overview

- **Four** distinct perspectives:

1. **Process** Quality: Targets the processes and **methods** to develop.
2. **Internal** Quality: Targets the **product's** software **artefacts**.
3. **External** Quality: Targets the **product's execution** and usage.
4. Quality in **Use**: Targets the **user** interacting with the product under specific context of usage.

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Quality Standards

- **External** Quality Requirements:
  - Derived from user **quality needs**.
  - Focus mostly on **functional** aspects.
  - … when software is being **executed**. (Availability, Mean time between failures, Latency)

- **Internal** Quality Requirements:
  - Based on **artifacts**: Static and dynamic models and **source code**.
  - Important from a **development perspective.** (Lines of code, coupling, cyclomatic complexity)
  - … should remain unchanged unless **redesigned.**

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Part 2:
Quality Attributes

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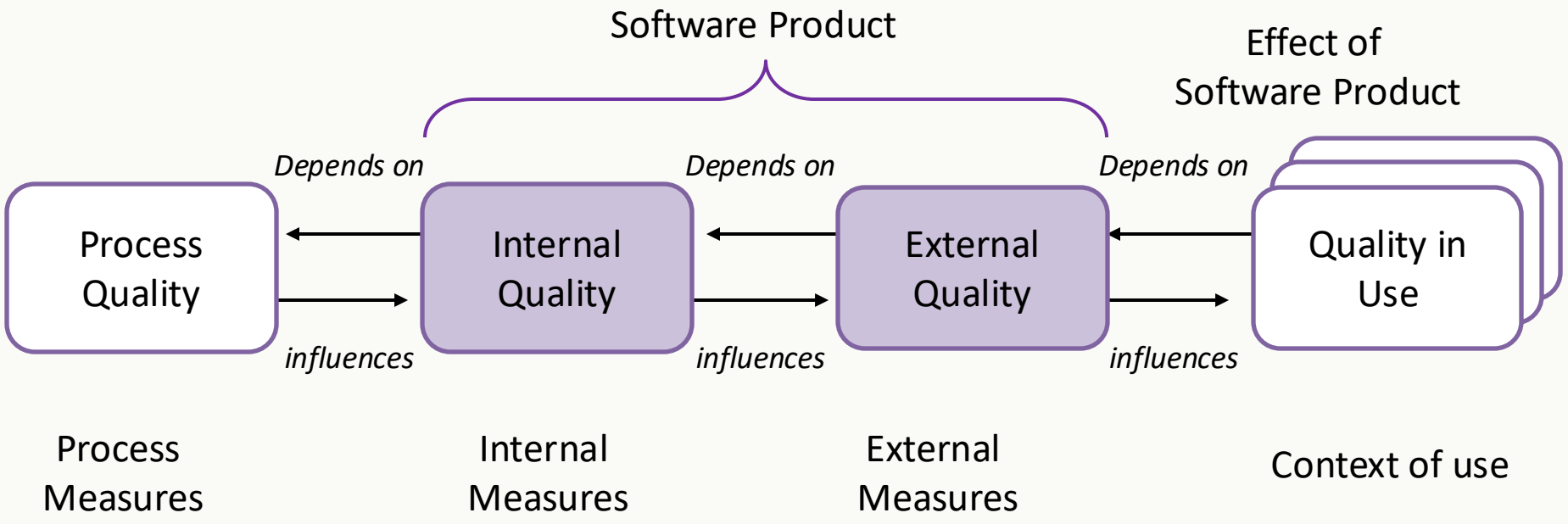
Quality Standards



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Product Quality

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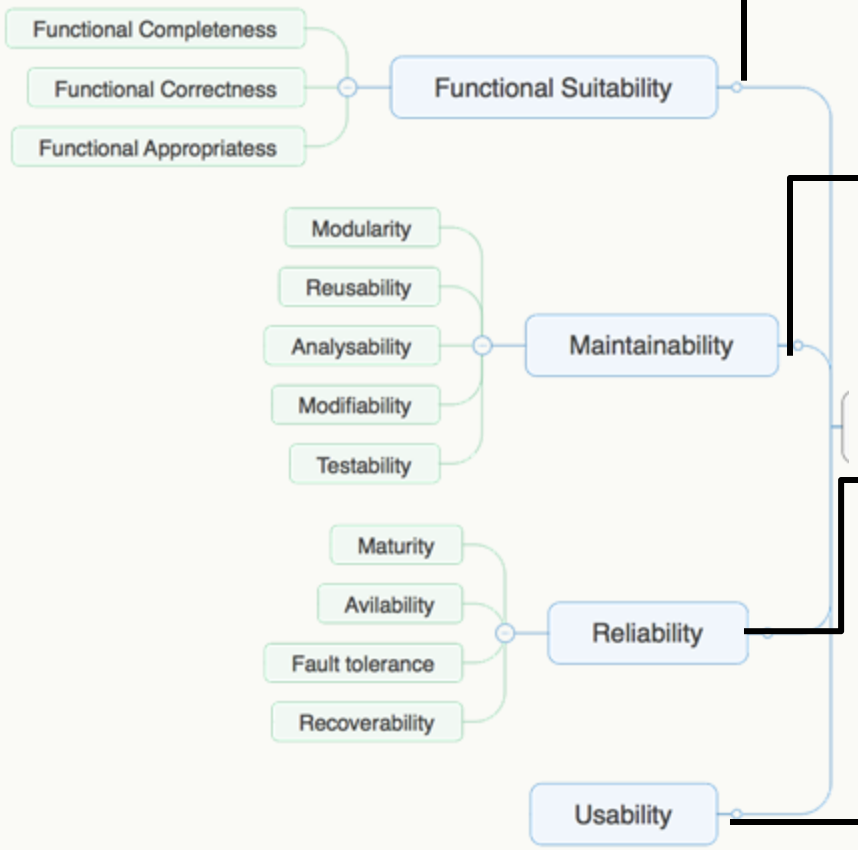
Product Quality



Product **meet stated and implied needs** when used under specified conditions

Product **can be modified** by the intended maintainers

Product **performs** specified functions **under** specified conditions **for** a specified period of time

Product can be **used by specified users** with satisfaction in a specified context of use

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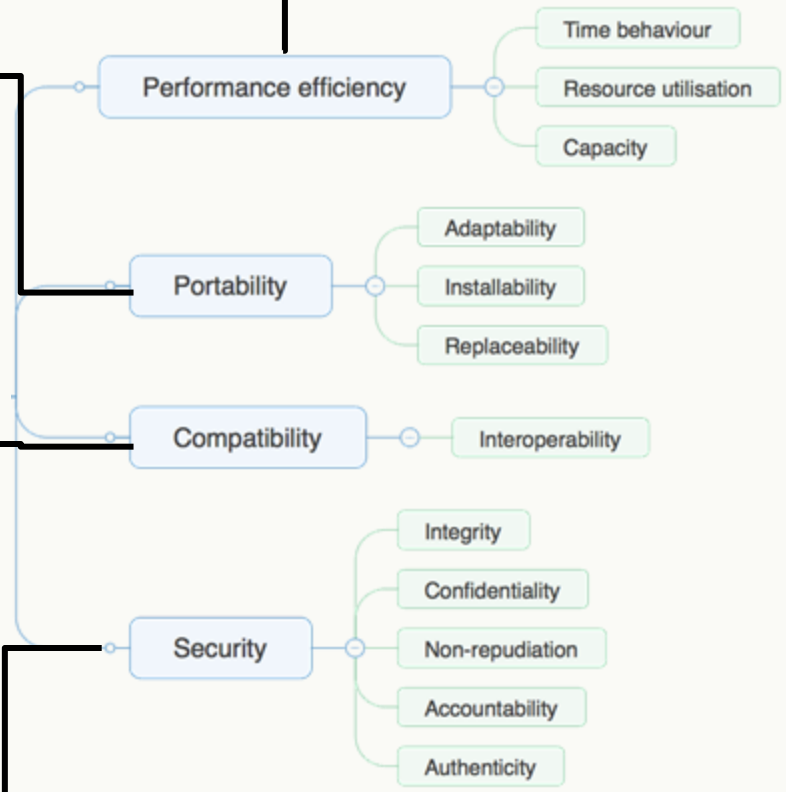
Performance relative to the **amount of resources** used under stated conditions

Product can be **transferred** from one hardware to another

Product **can exchange information**; Perform its required functions, while **sharing** the same hardware or software environment

Product **protects information and data** so that other have data access appropriate to their authorization level



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Let’s test our terminology…

- Go to mentimeter:

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Summary: Quality Attributes

- ISO 25010:2011: **7 quality attributes** + Functional Suitability
- Ideally, their definition should be **complete** and **non-overlapping**
- We use quality scenarios to define **measurable and testable** descriptions of quality attributes.

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